

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

Year 11

2 Unit

May 2006

Common Test

Time allowed: 70 mins

Name: _____

Teacher: _____

Instructions:

- Begin each question on a new page
- Marks shown are approximate and may be varied
- Show necessary working
- Full marks may not be awarded if working is poorly set out or difficult to read

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
/8	/7	/7	/7	/8	/7	/8	/6	/58

Question 1

- a) Evaluate $\frac{1.3 \times 10^{-3}}{8.6 - 4.2}$. Give your answer in scientific notation using 2 significant figures. 2
- b) Express $0.\dot{4}\dot{2}$ as a simplified fraction. 2
- c) Simplify $\sqrt{27} - \sqrt{12}$ 2
- d) Evaluate $x^2 - 3x + 1$ when $x = 3\sqrt{5}$. Leave your answer in simplest form 2

Question 2

- a) Simplify: (i) $3x - 4x(3 - x)$ 1
- (ii) $(4a^3b)^2 \times \frac{a^2}{6b^3}$ 2
- b) Express the answer to $\sqrt{x} \div \frac{1}{x}$ in simplified index form 2
- c) Express $8^a \times 4^{a+1}$ as a power of 2 2

Question 3

- a) Factorise (i) $x^2 - 5x - 6$ 1
- (ii) $x^3 - 8$ 1
- (iii) $x^3 + x^2 + x + 1$ 1
- b) Simplify (i) $\frac{x^2 - x}{x^2 - 1}$ 2
- (ii) $\frac{4}{x+2} - \frac{1}{x}$ 2

Question 4

- a) Evaluate $|-3| - |-4|^2$ 1
- b) Solve (i) $\frac{2x-4}{3} = x+2$ 2
- (ii) $2x^2 + 9x - 5 = 0$ 2
- (iii) $|1 - 2x| = 6$ 2

Question 5

- a) Solve simultaneously $3x - y = 0$ and $4x - 2y = 3$ 3
- b) Solve $\frac{4}{x+2} + x = 3$ 2
- c) Given $f(x) = 3x^2 - 2x$, find and simplify:
- (i) $f(-3)$ 1
- (ii) $f(x+h) + f(x)$ 2

Question 6

a) Sketch each function. Use a ruler for all straight lines and show any x, y intercepts:

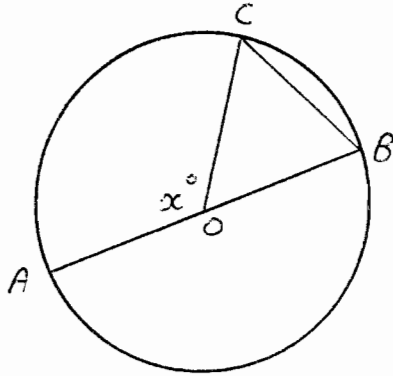
- (i) $y = \frac{1}{x+1}$ 1
- (ii) $y = -\sqrt{4-x^2}$ 1
- (iii) $y = |3x-6|$ 1
- b) (i) Sketch $y = x^2 + 2x - 3$. Show all intercepts 2
- (ii) Solve $x^2 + 2x - 3 \leq 0$. You may use the sketch above to help 1
- (iii) What is the range of the function in (i) 1

Question 7

- a) State the domain and range of the function $y = \sqrt{x-4}$ 2
- b) Find the centre of the circle $x^2 - 4x + 4 + y^2 + 6y = 0$ 2
- c) (i) Sketch $y = 2^x$ and $y = |x-1|$ on the same axes 2
- (ii) Hence or otherwise, solve $|x-1| > 2^x$ 1
- (iii) On your graph in (i), neatly shade the intersected region of $x \geq 0$ and $y \geq 2^x$ 1

Question 8

a)

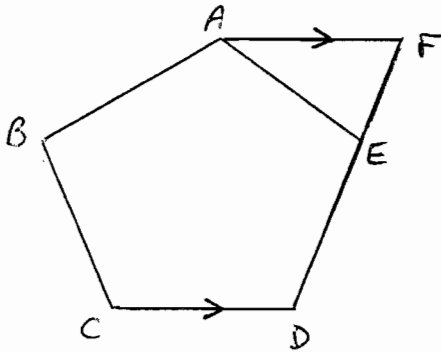


O is the centre of the circle. If $\angle AOC = x^\circ$, prove

that $\angle B = \frac{x^\circ}{2}$

3

b)



ABCDE is a regular pentagon. $AF \parallel CD$ and DE is produced to F.

Set out a geometrical proof to find the size of $\angle FAE$. 3

SOLUTIONS

① a) $3.0 \times 10^{-4} \leftarrow \textcircled{1}$

b) $x = 0.4242 \dots$

$100x = 42.4242 \dots$

$99x = 42 \quad \textcircled{1} \text{ for correct method}$

$x = \frac{42}{99}$

$= \frac{14}{33} \leftarrow \textcircled{1}$

① ↘

c) $3\sqrt{3} - 2\sqrt{3} = \sqrt{3} \leftarrow \textcircled{1}$

d) $(3\sqrt{5})^2 - 3(3\sqrt{5}) + 1$

$= 9 \times 5 - 9\sqrt{5} + 1 \quad \textcircled{1} \text{ for substitution}$

$= 46 - 9\sqrt{5} \leftarrow \textcircled{1}$

② a) (i) $3x - 12x + 4x^2$
 $= 4x^2 - 9x \quad \textcircled{1}$

(ii) $16a^6b^2 \times \frac{a^2}{6b^3}$
 $\textcircled{1}$

$= \frac{8a^8}{3b} \leftarrow \textcircled{1}$

b) $x^{\frac{1}{2}} \div x^{-1} = x^{\frac{3}{2}} \quad \textcircled{1}$

c) $(2^3)^n \times (2^2)^{n+1} \leftarrow \textcircled{1}$

$= 2^{3n} \times 2^{2n+2}$

③ a) (i) $(x-6)(x+1) \quad \textcircled{1}$

(ii) $(x-2)(x^2+2x+4) \quad \textcircled{1}$

(iii) $x^2(x+1) + 1(x+1)$

$= (x+1)(x^2+1) \quad \textcircled{1}$

b) (i) $\frac{x(\cancel{x-1})}{(x+1)(\cancel{x-1})} = \frac{x}{x+1} \quad \textcircled{1}$

(iii) $\frac{4x}{x(x+2)} - \frac{1(x+2)}{x(x+2)} \quad \textcircled{1}$

$= \frac{4x - x - 2}{x(x+2)}$

$= \frac{3x - 2}{x(x+2)} \quad \textcircled{1}$

④ a) $3 - 16 = -13 \quad \textcircled{1}$

b) (i) $2x - 4 = 3x + 6 \quad \textcircled{1}$

$-x = 10$

$x = -10 \quad \textcircled{1}$

(ii) $(2x-1)(x+5) = 0 \quad \textcircled{1}$

$\therefore x = \frac{1}{2}, -5 \quad \textcircled{1}$

(iii) $1 - 2x = 6 \text{ or } -(1 - 2x) = 6 \quad \textcircled{1}$

$-2x = 5 \text{ or } -1 + 2x = 6$

$x = -2\frac{1}{2}$

$2x = 7$

5 a) $3x - y = 0$ — (1)
 $4x - 2y = 3$ — (2)

(1) $\times -2$: $-6x + 2y = 0$ 1 mark for clear method

(2) + (3): $-2x = 3$

$x = -1\frac{1}{2}$ 1 mark

Sub in (1): $-4\frac{1}{2} - y = 0$

$y = -4\frac{1}{2}$ 1 mark

b) $4 + x(x+2) = 3(x+2)$

$4 + x^2 + 2x = 3x + 6$

$x^2 - x - 2 = 0$ ← (1)

$(x-2)(x+1) = 0$

$x = 2$ or -1 ← (1)

c) (i) $f(-3) = 27 + 6$

$= 33$ ← (1)

(ii) $3(x+h)^2 - 2(x+h) + 3x^2 - 2x$ ← (1)

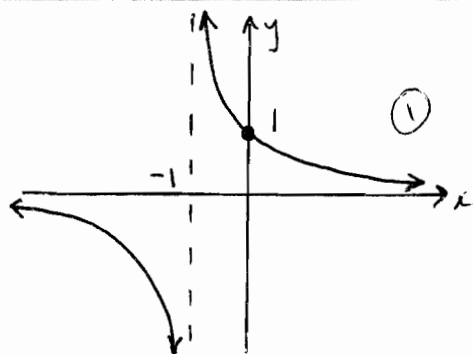
$= 3x^2 + 6xh + 3h^2 - 2x - 2h + 3x^2 - 2x$

$= 6x^2 + 6xh + 3h^2 - 4x - 2h$

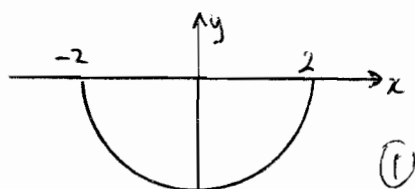
← (1)

(6) a) (i)

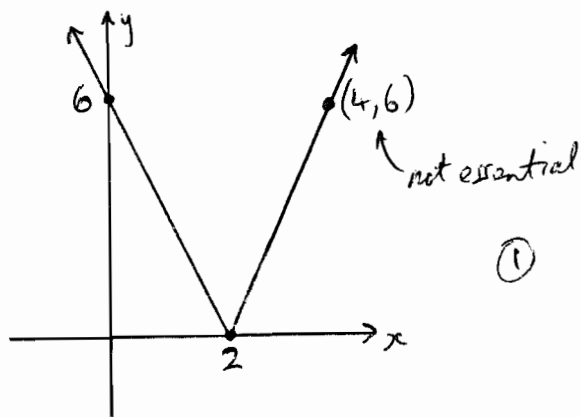
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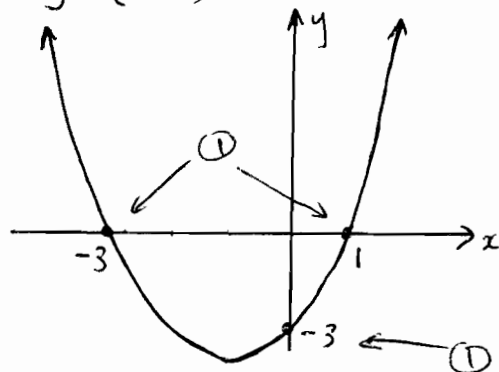
(ii)



(ii)



b) (i) $y = (x-1)(x+3)$



(ii) $-3 \leq x \leq 1$ (1)

(iii) min. value when $x = -1$
 $\therefore y = -4$

$\therefore$ range is $y \geq -4$

⑦ a) $x - 4 \geq 0$

$\therefore D: x \geq 4$ ①

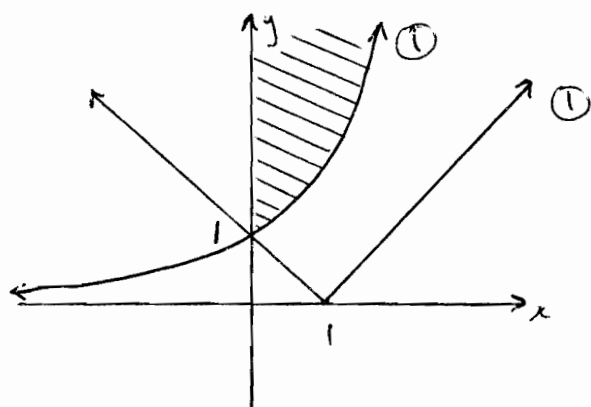
$R: y \geq 0$ ①

b) $(x-2)^2 + y^2 + 6y + 9 = 9 \leftarrow$ ①

$(x-2)^2 + (y+3)^2 = 9$

$\therefore$ centre at $(2, -3)$ $\leftarrow$ ①

c) (i)



(ii) $x < 0$ ①

(iii) above ①

⑧ a) $x = \angle B + \angle C$ (exterior angle of $\triangle OBC$)

$\triangle OBC$ is isosceles ($OC = OB$, equal radii)

$\therefore \angle B = \angle C$ (base angles)

$\therefore \angle B = \angle C = \frac{x}{2}^\circ$

$\leftarrow$ 1 mark deducted for each error of reasoning

b) $\angle D = 108^\circ$ (angle sum for reg. pentagon is 540°)

$\therefore \angle F = 72^\circ$ (co-interior angles, $AF \parallel CD$)

Similarly, $\angle DEA = 108^\circ$

Now, $\angle DEA = \angle F + \angle FAE$ (exterior angle rule for $\triangle FAE$)

$\therefore \angle FAE = 108^\circ - 72^\circ$
 $= 36^\circ$